

Insurance Loss Analytics: Predicting Loss Cost & Claim Likelihood

Building data-driven models to better price risk and improve profitability

Group 5

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DSO 530 · USC Marshall · Spring 2026

🎯 PROBLEM

Insurance pricing depends on accurate loss prediction.

Mispricing leads to adverse selection: low-risk customers leave for cheaper competitors, leaving the insurer with an unprofitable, high-risk pool.



📊 DATA INSIGHTS

🔄 88.85%
Zero-claim rate (only 11.15% have claims)

📈 Loss distribution is highly skewed
with a long right tail

📊 DATA ENGINEERING

28 raw variables to 33 features
Included actionable time features and behavioral features

⚙️ MODELING APPROACH

Two-part + Tweedie framework

Features
(33 engineered features)

CS Model
(Classification) Claim Likelihood

LC Model
(Tweedie Regression) Loss Severity

Expected Loss
= Claim Prob. × Loss Cost

🎯 Tweedie models handle zeros and heavy tails

🔄 3-fold CV on 80% train, evaluated on 20% hold-out

🛡️ No target leakage: X.15–X.18 excluded from features

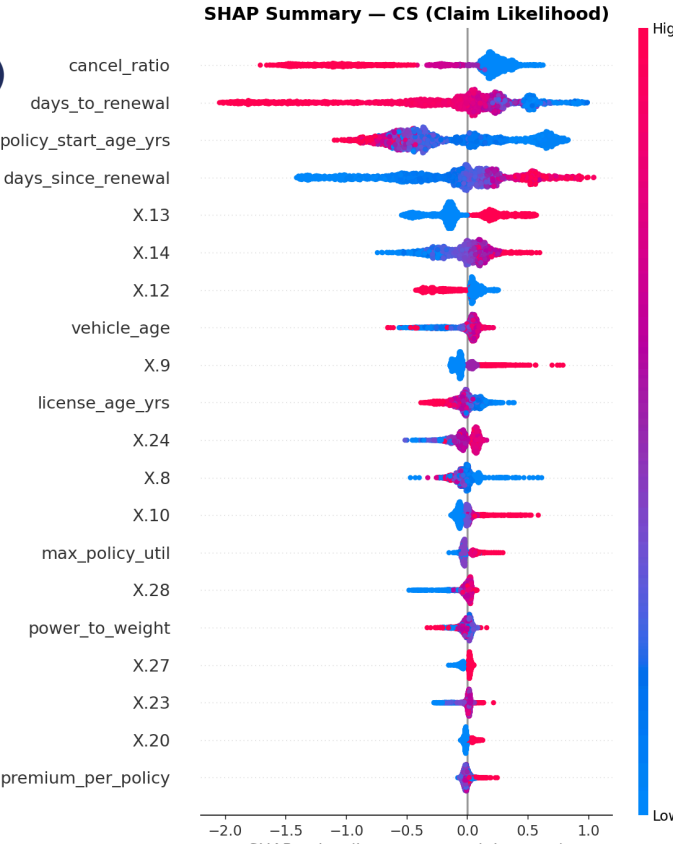
📈 MODEL PERFORMANCE

Target	Final Model	Validation Score	Why Chosen
LC (Loss Cost)	LightGBM (Tweedie)	MSE = 254,710	- Within 1% of best model - Consistent pipeline with HALC - Faster training & native categorical support
HALC (Hist. Adj. Loss Cost)	LightGBM (Tweedie)	MSE = 996,779	- Best validation MSE - Modeled directly since X.18 is unavailable at prediction time
CS (Claim Status)	XGBoost (Classifier)	AUC = 0.857	- Outperformed Logistic Regression & Random Forest - scale_pos_weight handles class imbalance

🔍 KEY BUSINESS INSIGHTS FROM SHAP ANALYSIS

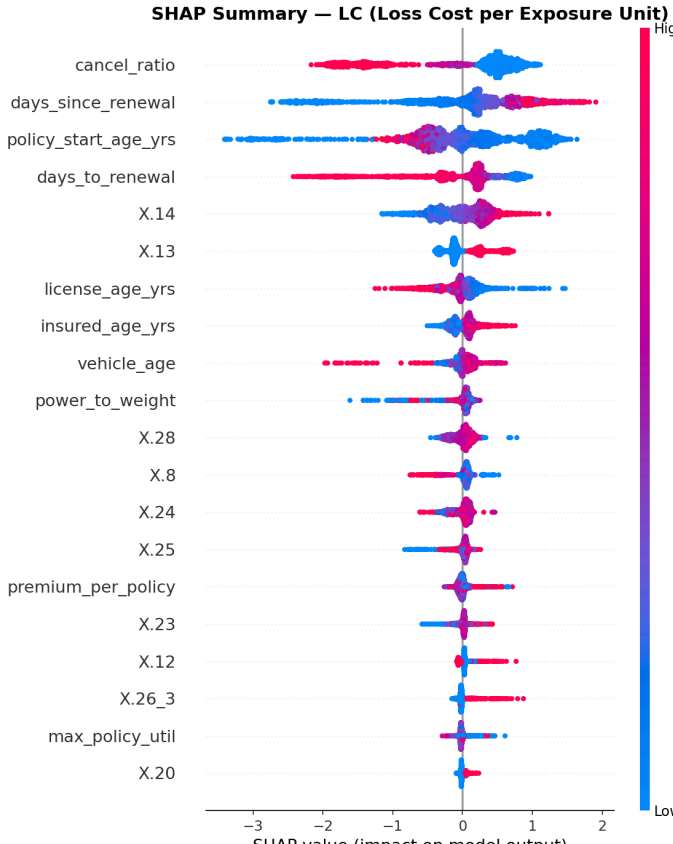
Claim Status (CS) — Top Features

Behavioral & renewal-related features are the strongest drivers of claim likelihood.



Loss Cost (LC) — Top Features

Cancellation behavior and renewal timing are key drivers of loss cost.



Consistent Pattern Across Models

- Behavioral signals dominate** risk across all models, far outweighing demographics.
- Risk increases near renewal** periods, suggesting potential strategic claiming behavior.
- Premium** captures only part of risk; additional behavioral signals provide significant predictive power.
- Six of top eight features overlap** across LC, HALC, and CS — a robustness signal.

What This Means for the Business

- ✓ Incorporate behavioral variables (cancellation history, renewal timing) into pricing models.
- ✓ Focus on policies near renewal for underwriting review and engagement.
- ✓ Move beyond premium-only pricing to reduce adverse selection and improve profitability.

➔ Pricing should incorporate customer behavior, not just demographics.

TAKEAWAY: Behavior > Demographics

The dominant risk signal is not who the customer is, but how they act: cancellation history, renewal timing, tenure outrank age, vehicle type, and premium. Actionable for retention pricing, renewal strategy, and underwriting policy, but not transferable to life insurance.